Shubhkaran Dhillon

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Experience

FastTrack Technologies Inc. Mar 2026 – Present
Software Engineer *Edmonton, AB*

- Own end-to-end **backend and cloud infrastructure** for a production **LMS and examination platform** serving public universities and **150K+ users**, including incident response, capacity planning, and architecture decisions.
- Built cloud-native infrastructure on **AWS** (**S3**, **ECS**, **RDS**, **ElastiCache**), achieving **99.9% uptime**, improving performance by **30%**, and reducing latency by **25%**.
- Built a **CI/CD failure triage agent** using **LangGraph** that autonomously investigates failed pipeline runs, classifies root cause, and posts structured summaries to **Slack**, reducing mean debugging time by **70%**.
- Implemented **Infrastructure as Code** using **Terraform**, reducing manual configuration errors by **90%** and cutting environment provisioning time by **60%**.
- Led **cloud cost optimization** initiatives across **AWS** services, reducing monthly infrastructure spend by **20%** through right-sizing, auto-scaling, and resource cleanup.

Cardify Jan 2025 – Aug 2025
Software Engineer Intern *Remote*

- Built a full-stack **digital business card** platform using **Spring Boot**, **React**, and **PostgreSQL**, onboarding **1,000+ users** during pilot testing.
- Designed an **AI agent** to generate optimized profile content from user inputs, reducing onboarding time by **35%** and increasing profile completion rate by **40%**.
- Optimized **PostgreSQL** schema and queries through indexing and restructuring, improving database efficiency by **25%**.

Projects

Learning Activity Games | *Django, React.js, LangChain, OpenAI, MongoDB*

- Built an AI-powered learning platform deployed by the **University of Alberta** Psychology Department for **2,000+ students**, generating Jeopardy-style quizzes from lecture content using **LangChain** and **OpenAI**, improving active recall and classroom engagement by **40%**.

QR Code Event Check-In | *Java, Firebase, Espresso, JUnit, Mockito, Maven*

- Led a team of **6** to build an **Android** event check-in platform for a client, enabling QR code-based attendee verification and real-time tracking across **10+ events** and **500+ attendees**; reduced manual check-in time by **60%** through **Firebase** real-time sync and automated notifications.

Education

University of Alberta June 2025
Bachelor of Science in Computer Science, Minor in Economics *Edmonton, AB*

- **Relevant Coursework:** Machine Learning, Software Engineering, Artificial Intelligence, Data Structures & Algorithms, Computer Networks, Distributed Systems, Computer Security, Database Management

Certifications

- AWS Certified Developer Associate - Amazon Web Services (AWS)
- AWS Certified Cloud Practitioner - Amazon Web Services (AWS)
- Generative AI for Software Development - DeepLearning.AI

Technical Skills

Languages: Java, Python, JavaScript/TypeScript, SQL
Backend: Spring Boot, Django, Node.js, REST APIs, Microservices
AWS & Cloud Architecture: EC2, S3, Lambda, RDS, SQS, ECS, VPC, IAM, Multi-AZ/HA, Auto Scaling
DevOps: Docker, Kubernetes, Terraform, CI/CD (GitHub Actions)
Databases & Storage: PostgreSQL, MySQL, MongoDB, Firebase
Distributed Systems: Redis (caching), Kafka (event streaming), Load Balancing, Horizontal Scaling
Frontend: React.js, HTML/CSS
Core Concepts: DS&A, OOP, System Design, Networking, Database